

Assignment 5

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Task 1

Task 1.a

We estimate the following regression:

Model (1):

$$\ln(\text{medexp}) = \beta_1 + \beta_2 \text{privateins} + \beta_3 \text{phylum} + \beta_4 \text{actlim} + \beta_5 \text{totchr} + \beta_6 \text{age} + \beta_7 \text{female} + \varepsilon \quad (1)$$

And get the following result:

. reg lnmedexp privateins phylim actlim totchr age female						
Source	SS	df	MS	Number of obs	=	2,955
				F(6, 2948)	=	144.51
Model	1255.67684	6	209.279474	Prob > F	=	0.0000
Residual	4269.21254	2,948	1.4481725	R-squared	=	0.2273
				Adj R-squared	=	0.2257
Total	5524.88938	2,954	1.87030785	Root MSE	=	1.2034
lnmedexp	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
privateins	.2753629	.0455897	6.04	0.000	.185972	.3647539
phylim	.3009026	.0570198	5.28	0.000	.1891	.4127053
actlim	.3444701	.0619955	5.56	0.000	.2229111	.466029
totchr	.3742093	.0184279	20.31	0.000	.3380765	.4103421
age	.0029146	.0036421	0.80	0.424	-.0042267	.010056
female	-.096344	.0453305	-2.13	0.034	-.1852266	-.0074614
_cons	6.829525	.2723953	25.07	0.000	6.295421	7.36363

Figure 1: Results of the regression

Task 1.b

$$SE(\hat{\beta}_2) = \sqrt{\hat{\sigma}^2 \cdot (X^T X)^{-1}_{22}} = \sqrt{\frac{RSS}{N-K} \cdot (X^T X)^{-1}_{22}} = \sqrt{\frac{4269.2955-7}{2955-7} \cdot (X^T X)^{-1}_{22}} = \sqrt{1.448 \cdot (X^T X)^{-1}_{22}}$$

Task 1.c

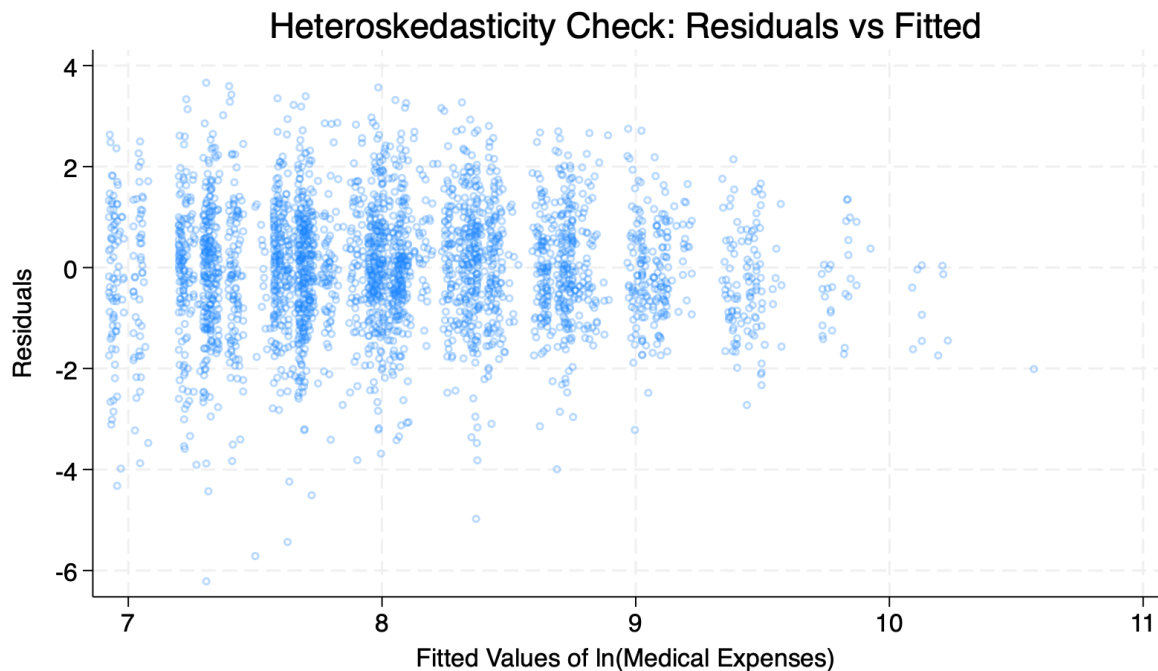
β_2 captures the percentage difference in medical expenses between people who have private medical insurance and people who do not. The change in the outcome is measured in percentage points, because of the logarithm (let medexp = Y, privateins = X):

$$d\ln(Y)/dX = \frac{1}{Y} \left(\frac{dY}{dX} \right) = \beta_2 \text{ therefore } \frac{dY}{Y} = \beta_2 \cdot dX$$

Task 1.d

To check whether we are dealing with heteroskedastic disturbances, we would plot the fitted values of the outcome variable against the residuals. The homoskedasticity assumption states that disturbances have the same variance under different values of X. The predicted value of Y is the linear combination of X variables, so the errors should have approximately the same variance at different level of Y predicted under homoskedasticity.

Here is the plot:



It seems to us that some heteroskedacity is possible in this case, because for the higher level of the predicted outcome the variance of the errors seems lower than for the lower level of outcome. Therefore, there may be some unobservable factors that are correlated with the outcome. However, it should be further tested.

Task 1.e

Task 1.e.i

Here are the results we get from doing two steps from the assignment:

It clearly shows the presence of heteroskedacity.

```

. display "BP = " %9.4f BP
BP =    32.4256

. display "p-value = " %9.4f chi2tail(1, BP)
p-value =    0.0000

```

Task 1.e.ii

Here are the results we get from "estat hettest , iid" command:

```

. estat hettest , iid

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variable: Fitted values of lnmedexp

H0: Constant variance

      chi2(1) =    32.43
Prob > chi2 =    0.0000

.
end of do-file
.

```

The BP statistic is the same as in the previous exercise.

Task 1.e.iii

The output of the BP test was partly expected. We anticipated some level of heteroskedacity. However, the p-value seems very low, representing strong signs of heteroskedacity that we have not noticed.

Task 1.f

It seems that we have to adjust the calculation of standard errors, because currently standard errors are computed under homoskedacity assumption that clearly does not hold in this case. So, the p-values and confidence intervals are incorrectly computed, therefore we have to compute heteroskedacity standard errors.

Task 1.g

Here is the stata output with heteroskedacity-robust standard errors:

Task 1.h

R^2 shouldn't change when moving from OLS estimation to robust OLS, because with this change the coefficients itself do not change. Therefore, fitted values and errors do not change, and so RSS and TSS do not change. Therefore, $R^2 = 1 - \frac{RSS}{TSS}$ does not change.

```
. reg lnmedexp privateins phylim actlim totchr age female, robust
```

Linear regression	Number of obs	=	2,955
	F(6, 2948)	=	147.63
	Prob > F	=	0.0000
	R-squared	=	0.2273
	Root MSE	=	1.2034

lnmedexp	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
privateins	.2753629	.0463625	5.94	0.000	.1844567	.3662692
phylim	.3009026	.0576555	5.22	0.000	.1878535	.4139518
actlim	.3444701	.0630747	5.46	0.000	.2207952	.4681449
totchr	.3742093	.0187345	19.97	0.000	.3374753	.4109433
age	.0029146	.0036938	0.79	0.430	-.0043281	.0101573
female	-.096344	.0455283	-2.12	0.034	-.1856145	-.0070735
_cons	6.829525	.2787707	24.50	0.000	6.28292	7.37613

Task 1.i